

Ideal_LTSpice_Netlist_2

Prompt

What should i plot and observe ?

Answer

In LTSpice, after running the simulation, you should plot the voltages at specific nodes to follow the signal's journey from the antenna to your headphones.

Here is exactly what you should plot to see the radio working step-by-step:

- **V(ANT) (The Raw Signal):** This shows the incoming AM signal from the antenna. You will see a dense 693 kHz carrier wave whose "envelope" (the outer shape) smoothly goes up and down at 1 kHz.
- **V(RF_C) (The RF Amplification):** This is the output of your first transistor stage. It will look identical in shape to the antenna signal, but the amplitude will be significantly boosted.
- **V(DET_OUT) (The Demodulation):** This is the output of your diode detector. You will see that the dense, high-frequency carrier wave has been stripped away. What remains is the recovered 1 kHz audio sine wave riding on a small DC voltage offset.
- **V(OUT) (The Final Audio):** This is the final signal after the audio amplifier and the output coupling capacitor. It will be a clean, boosted 1 kHz sine wave centered exactly around 0V, which is what actually drives the 32 Ω headphones.

How to do it in LTSpice:

If you built the schematic visually, simply hover your mouse over the corresponding wires until the cursor turns into a red probe, then click. If you are running the text netlist directly, right-click the blank plot window, select "**Add Trace**", and choose those node names from the list.

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